

25. (1.75 points)

a. (0.5 point)

Describe a behavioral approach to modeling the underwriting cycle.

b. (1.25 points)

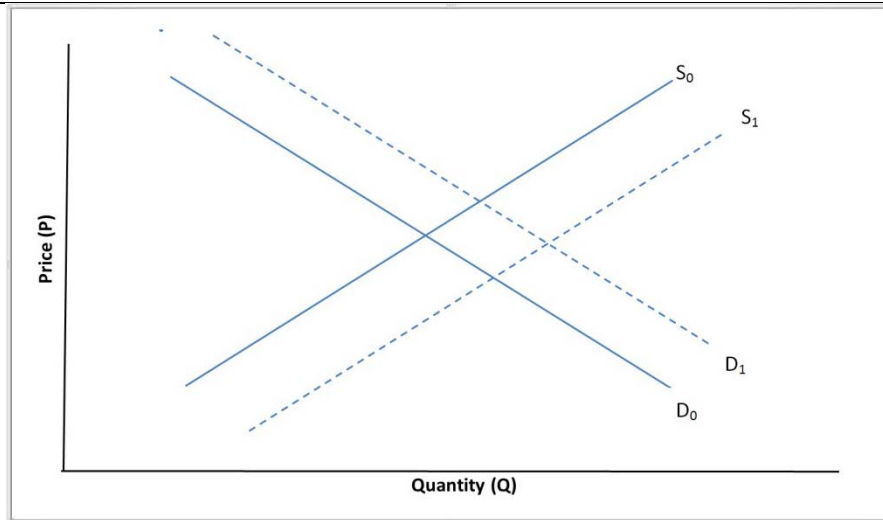
Graph a supply and demand curve showing how an industry-wide capital infusion would shift the curves. Briefly describe the rationale behind the shifts.

END OF EXAMINATION

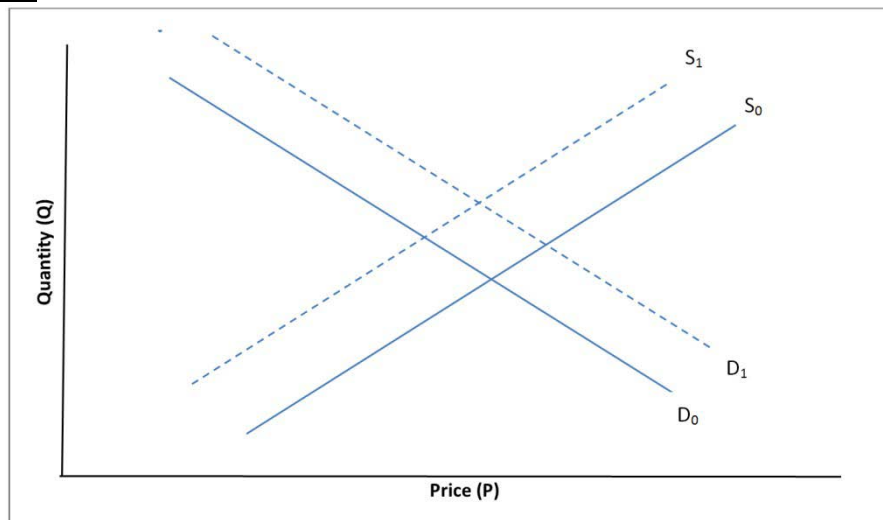
EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 25	
TOTAL POINT VALUE: 1.75	LEARNING OBJECTIVE(S): C8
SAMPLE ANSWERS	
Part a: 0.5 point	
<u>Sample 1</u> Behavioral approach – a middle ground between soft and technical approaches wherein several individual components of the process are built upon economic and behavioral theory. <u>Sample 2</u> Econometric modeling: uses a combination of soft and technical methods to understand behavior drivers of cycle and the impacts on supply and demand. <u>Sample 3</u> In between soft and technical approach. Model change in supply and demand in insurance, capital flows, etc. to model equilibrium price. <u>Sample 4</u> A behavioral approach combines elements from soft (human element) and technical approaches (mathematical rigor). It uses supply and demand to look at equilibrium prices. Adjust certain key inputs such as shock events, amount of capital, inflation, etc. which will influence the curves. Going through many iterations of the effect on the supply and demand curves creates a model of the underwriting cycle. <u>Sample 5</u> Econometric modeling is a behavioral approach to modeling the underwriting cycle. It incorporates mathematical formulas and rigor as well as the impact of human factors. By incorporating various components such as supply and demand curves, capital flows, etc. we are able to vary these assumptions to determine their impact on equilibrium prices and create an empirical distribution of potential future equilibrium prices, which can then be used to predict turns in the UW cycle.	
Part b: 1.25 points	
<u>Sample Graph 1</u>	

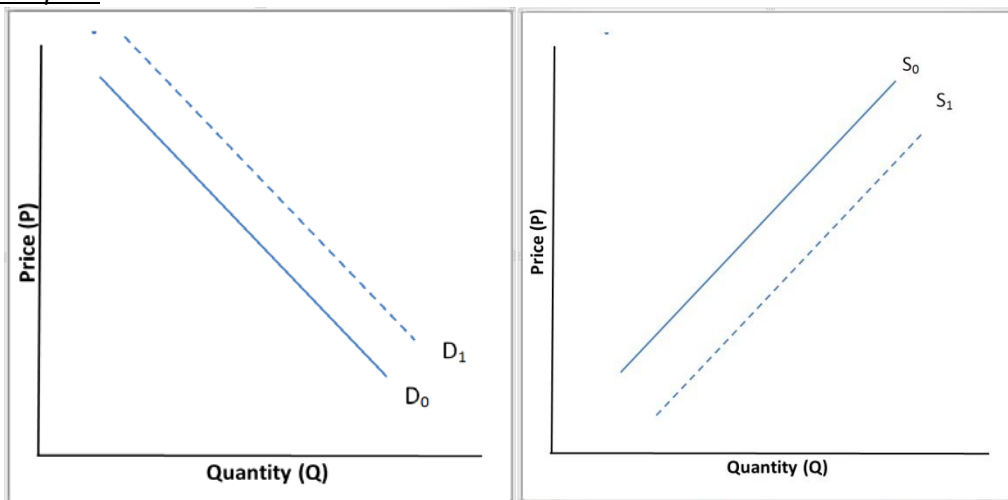
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Sample Graph 2



Sample Graph 3



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Sample Answers for the rationale behind the supply curve shift:

- Insurers will have increased capacity.
- Insurers will be able to take on more risk while remaining at the same probability of default.
- Insurers will be able to rely less on premium to provide them with capital and they can lower the price.
- Insurers will have more capital available to expand.
- Insurers will be willing to write more business.
- Insurers will be able to write more business.
- There will be more new entrants, increasing competition and driving prices down.
- More capital will lead to technical advancement; expense reduced.

Sample Answers for the rationale behind the demand curve shift:

- Insurance is worth more to the consumer because the insurer has a smaller probability of default.
- Consumers recognize the improved quality of the product.
- The insurers' promise is worth more because they have more money to pay claims.
- The insurers offer more protection.
- Consumers prefer well-capitalized insurers.
- Consumers prefer stable insurers.
- Consumers have more confidence in the insurance industry.
- A capital infusion increases an insurer's quality/capital quality.

EXAMINER'S REPORT

Candidates were expected to describe the behavioral approach to modeling the underwriting cycle and how it relates to the soft and technical approaches, as well as demonstrate understanding of the supply and demand component of the behavioral model by showing how and why the curves shift after a capital infusion.

Part a

Candidates were expected to demonstrate understanding of how behavioral approaches compare to soft and technical approaches, as well as give a high-level explanation of either how a behavioral model works or the motivation behind the behavioral model.

Common errors include:

- Providing insufficient explanations of the behavioral model.
- Describing soft approaches (Delphi method and competitor analysis) to modeling the underwriting cycle rather than behavioral approaches.

Part b

Candidates were expected to draw the supply curve shift and the demand curve shift on a correctly labeled graph (the equilibrium price could either be higher or lower, but the equilibrium quantity must be higher), and then describe the reasons behind each shift.

Common errors include:

- Drawing the supply and demand curves at the wrong angles, or drawing the shifts in the wrong directions. Some candidates struggled with the graph even when their verbal

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explanations were correct.

- Not showing any change to the demand curve.
- Not providing the rationale behind the shifts. Many candidates simply re-stated the direction of the shifts, without explaining why they occurred.
- Labeling the axes incorrectly.